

CS & IT ENGINEERING



Computer Network

Network Delays

Lecture No. - 10



By - Abhishek Sir



Recap of Previous Lecture



Topic

Basics of Switching





Topics to be Covered



Topic

Network Delays



* Delays in Computer Network



Topic : Network Delays



-> Four types of network delays :

1. Transmission delay
2. Propagation delay
3. Queuing delay
4. Processing delay



Topic : Transmission Delay

→ Time required to transmit a packet/frame over a link

$T_x \rightarrow$ Transmitter

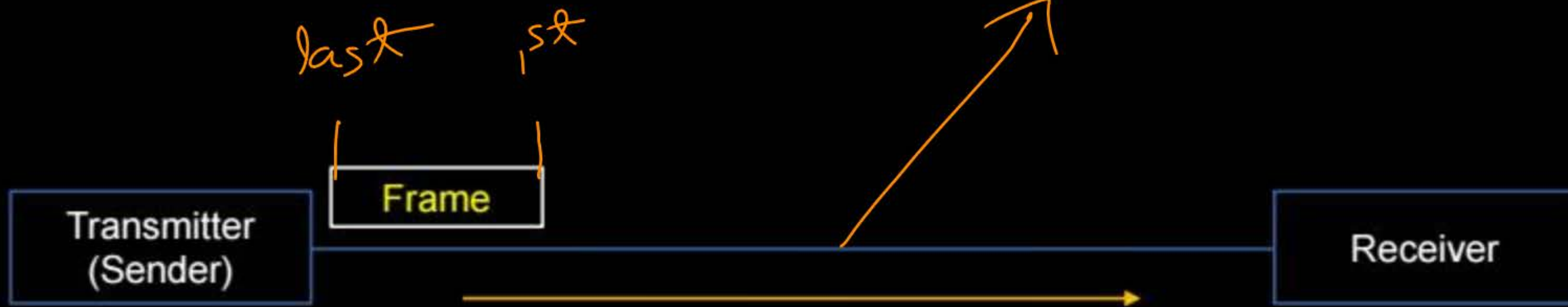
$R_x \rightarrow$ Receiver

→ Transmission Time / Delay $[t_x]$ (in seconds)



Topic : Transmission Delay

[Point-to-Point
Link]





Topic : Transmission Delay



$$\text{Transmission Delay } [t_x] = \frac{\text{Packet Size}}{\text{Data Transfer Rate}}$$



Topic : Propagation Delay

→ Time required to travel a **signal** (**bit**) from one end to other end of a link

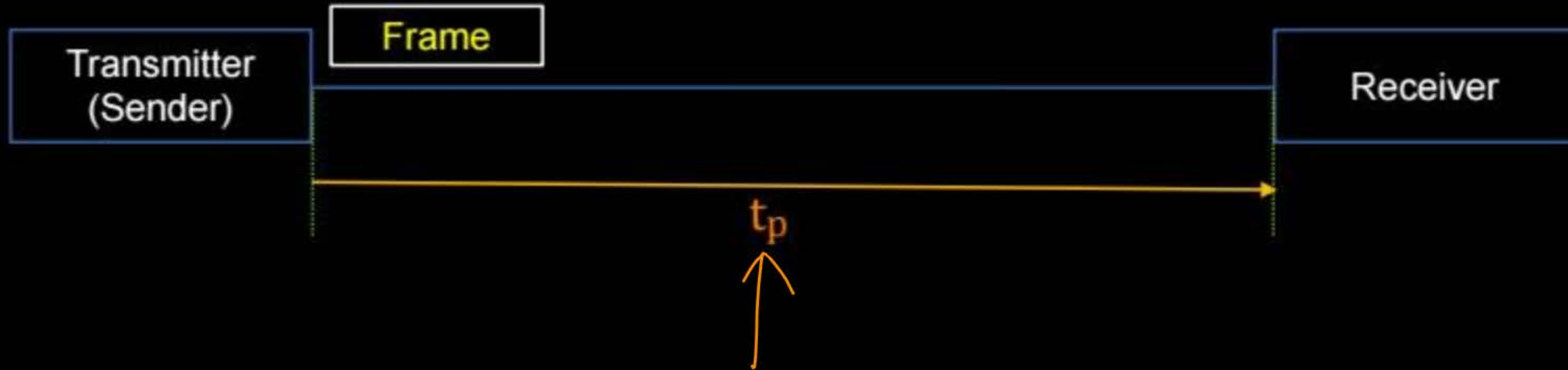
→ **Propagation Time / Delay** [t_p]

(in seconds)

→ **One-way** (by default)



Topic : Propagation Delay





Topic : Propagation Delay



→ if signal speed given in "meter per second"

$$\text{Propagation Delay} = \frac{[\text{Distance}]}{\text{Signal Speed}}$$



Topic : Propagation Delay



→ if signal speed given in "second per meter"

$$\text{Propagation Delay} = \text{Distance} \times \text{Signal Speed}$$



Topic : End-to-End Delay





Topic : End-to-End Delay

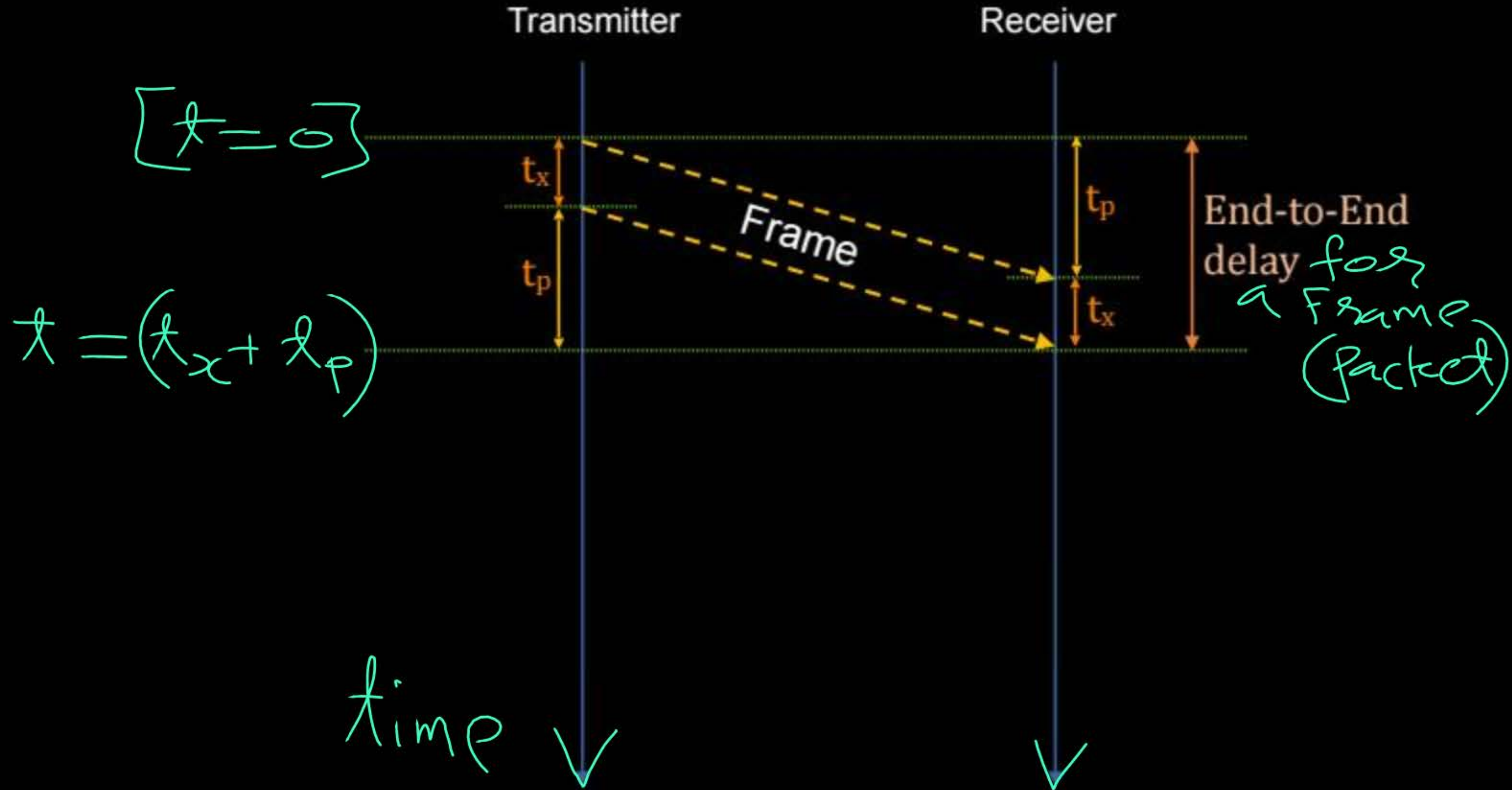
→ Time required for a packet to be transmitted from Transmitter to Receiver

→ One-way delay

(Frame)



Topic : End-to-End Delay





Topic : End-to-End Delay



End-to-end delay = Transmission delay + Propagation delay

$$= (t_x + t_p)$$

CS-17

$$t_x = P = ?$$

$$t_p = q = ?$$

CS-22

$$t_x + t_p = ?$$



Topic : Store-and-Forward Device

- Layer-2 device : 'Bridge' or 'Switch'
- Layer-3 device : 'Router'
- Layer-7 : Gateway

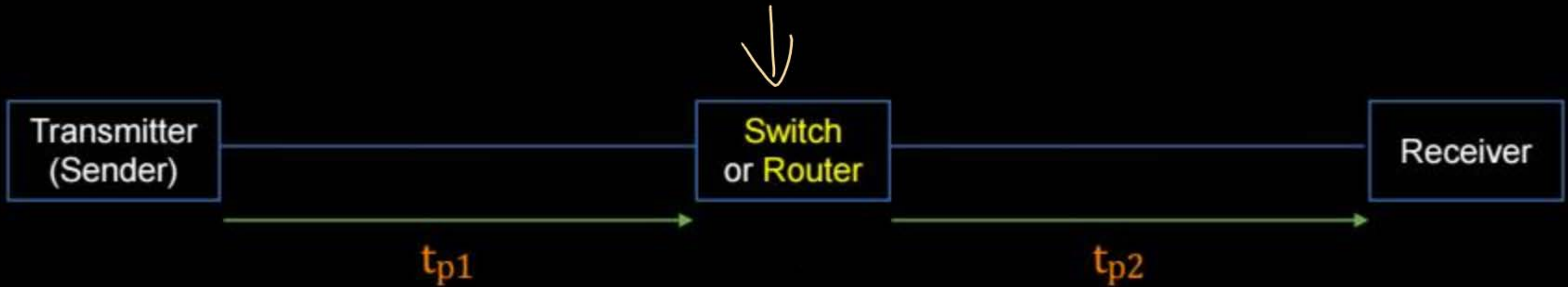
(Store-and-Forward Device)
⇒ store-and-forward delay

- Layer-1 (Physical layer)
* Devices : Repeater
Hub

(No memory)
Not store and forward device
* one-bit delay



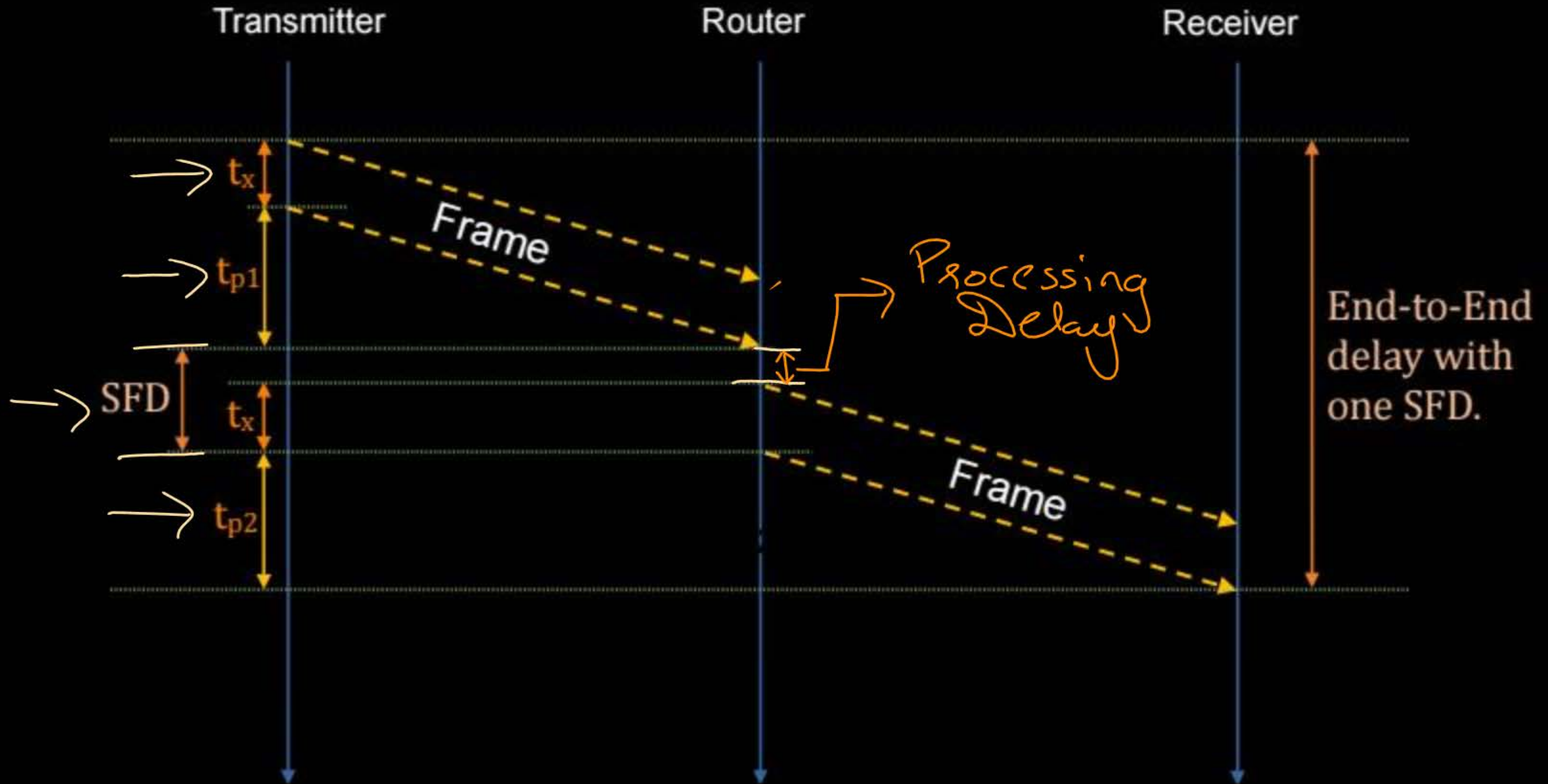
Topic : Store-and-Forward Device



→ Suppose all links have uniform bandwidth



Topic : End-to-End Delay





Topic : End-to-End Delay



- * For a Packet
- * Uniform bandwidth

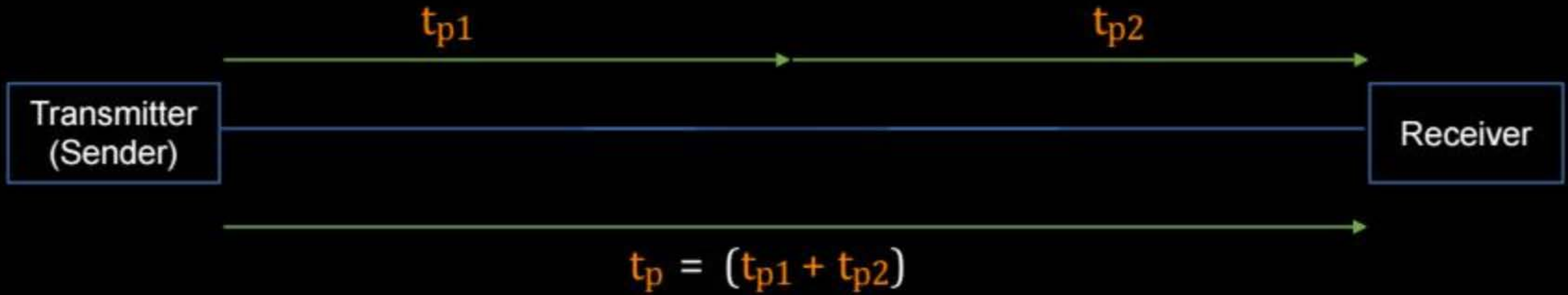
→ With one 'Store-and-Forward device'

$$\text{End-to-end delay} = \text{Transmission delay} + \text{Propagation delay for first link} \\ + \text{Store-and-Forward delay} \\ + \text{Propagation delay for second link}$$

$$= (t_x + t_{p_1}) + \text{SFD} + t_{p_2}$$



Topic : End-to-End Delay





Topic : End-to-End Delay



→ With no any 'Store-and-Forward device'

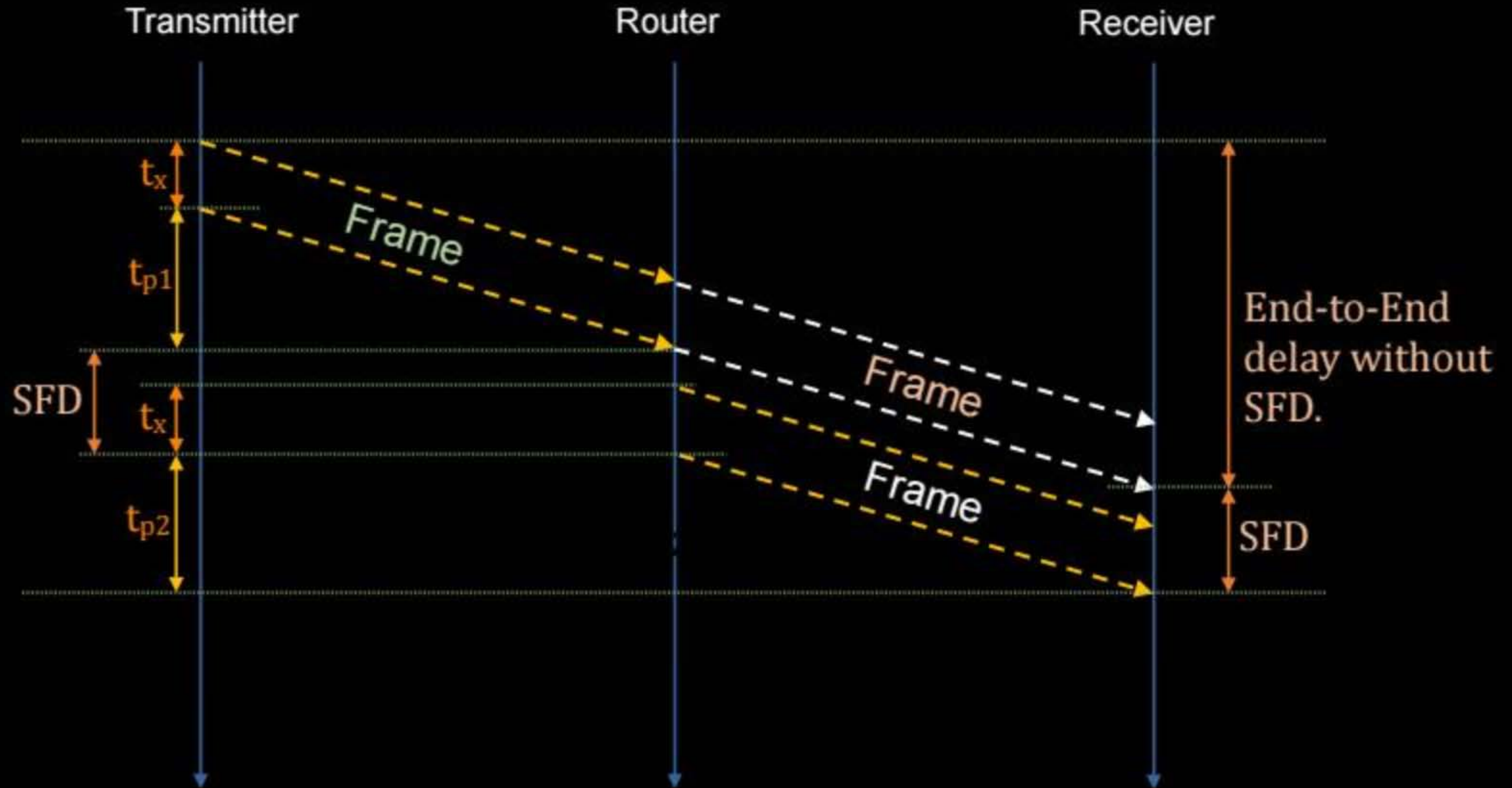
$$\text{End-to-end delay} = \text{Transmission delay} + \text{Propagation delay} = (t_x + t_p)$$

$$\text{End-to-End delay} = \text{Transmission delay} + \text{Propagation delay for first link} + \text{Propagation delay for second link}$$

$$= t_x + (t_{p_1} + t_{p_2})$$



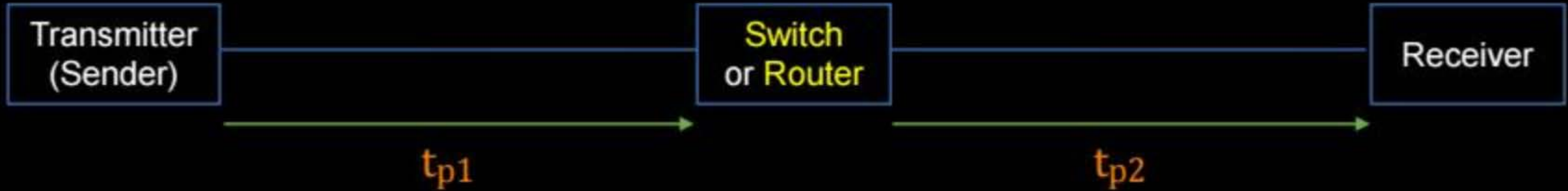
Topic : End-to-End Delay





Topic : Store-and-Forward Delay

(is t_x)



Store-and-Forward delay = Queuing delay at device + Processing delay by device
+ Transmission delay



2 mins Summary



Topic

Network Delays





THANK - YOU